

Role 1: Data Ingestion Technical Documentation

Project: Automated Airfare Index Pipeline

Scope: Role 1 Data Ingestion Specialist

Status: Verified & Locked (Internal Hackathon - Sept 3)

1. Architectural Overview

Role 1 serves as the data ingestion layer of the system. It handles automated target query matrix generation, maintains browser stealth to bypass bot detection, dynamically intercepts network XHR/Fetch API responses, and stages raw fare payloads into an append-only JSON Lines (JSONL) file.

2. Module Breakdown

- `src/ingestion/matrix_generator.py`: Generates dynamic target matrices for DEL-BOM across T+1 to T+5 advance booking windows using relative datetime offsets.
- `src/ingestion/stealth_browser.py`: Configures Playwright Chromium with anti-bot evasion arguments, suppresses navigator.webdriver flags, and rotates desktop User-Agents.
- `src/ingestion/scrapper_lead.py`: Listens to background HTTP responses, intercepts raw flight fare JSON payloads, and appends records to `seed_data/staging_raw_payloads.jsonl`.

3. Staging Output Contract

Output File: `seed_data/staging_raw_payloads.jsonl`

Schema Envelope: Contains `scraped_at` timestamp, `route_code`, `origin`, `destination`, `advance_window`, `departure_date`, `intercepted_url`, and `raw_payload` (`carrier`, `flight_no`, `total_quote`, `fare_details_html`, `api_response`).